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Review Article

On the current research progress of metallic materials fabricated by laser powder bed fusion process: a review



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ABSTRACT

Laser powder bed fusion (LPBF) is the most common metal additive manufacturing technique. Following pre-programmed designs, it employs a high-power density laser source to melt pre-alloyed or mixed powders layer by layer, allowing for complex metallic component fabrication.

This technique has recently been utilised to produce superior, near-full-density three-dimensional functional parts for various industrial applications. As the LPBF technology matures, ongoing research is being conducted to increase its viability as a sustainable solution in achieving digital transformation in metallic materials and qualifying new metallic materials for digital products. This review focuses on recent developments in the LPBF technique in terms of process parameters, defects, microstructure evolution, related metallurgical phenomena, and microselective laser melting processing for miniaturised part production. First, considerable attention is given to the related parameters that affect the LPBF process, that is, powder-related and laser-related properties. Second, the metallurgical imperfections related to the LPBF products are described in terms of their types, formation mechanisms, and suppression strategies for these defects. Third, the

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solidification behaviour, phase transformation, and precipitation during the LPBF processing were systematically investigated. Fourth, the materials implemented in micro-selective laser melting for three-dimensional microfeature production on various metals are summarised. Finally, the results from this review are summarized, and future research addressing existing difficulties and promoting technical advancements are recommended.

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1. Introduction

Additive manufacturing (AM), colloquially known as three-dimensional (3D) printing, offers great potential and exciting new opportunities for rapid prototyping of complex/individual parts of customised products in lesser time and with greater precision than conventional processes [1–4]. Recently, AM has skyrocketed to the point where end-use parts are being built for various functional applications, such as biomedicine, engineering, automotive, and aerospace [5–11]. Moreover, it holds many advantages, such as reduced tool costs, material resource savings, enhanced part reliability, design freedom, and complex structure manufacturing [12–15]. AM is a revolutionary technology that transforms tangible products into intangible digital content by introducing the concept of dematerialisation. There is a growing consensus that digitalised manufacturing profoundly affects business model innovation and society [16–18]. AM is considered a disruptive technology in the manufacturing sector [19] and has the potential to replace the economies-of-scale production model, which has lesser fixed costs than in economies-of-one production. As AM and 3D printing become more available, changes in business strategies, production processes, supply chains, logistics, and product life cycles are expected [20–24]. Therefore, existing companies and businesses must understand and realise the challenges and change their strategies and operations accordingly. The incredible developments in AM over the last years gave rise to ‘**digiproneurship**’ or the transformation of digital to physical **product entrepreneurship**, a term coined by Fox and Stucke [25]. According to Gibson et al. [26], ‘there is no fundamental reason for a product to be brought to markets through centralised development, production, and distribution. Instead, a product can be brought to markets through product conceptualization, product creation, and product propagation being carried out by individuals and communities in any geographical region.’ This paradigm shift in manufacturing technology will influence the way manufacturers think and consumers behave [27]. Petrick et al. [19] expressed that ‘manufacturing digiproneurship will allow anyone to return to the garage and make things that satisfy the needs of one or a very few customers.’ This emphasises the widespread use of AM technology and the revolution in the way products are manufactured and dispatched.

Laser powder bed fusion (LPBF) in additive manufacturing technology for metallic materials has fascinated researchers. This 3D printing process is applied using a selective laser melting (SLM) machine. This technique provides enormous

advantages over conventional manufacturing routes. For example, fabricating complex-shaped parts without further post-processing and almost no geometric restrictions are possible. The technique offers high material usage, high flexibility, and low production time. Moreover, high-density, near-net-shape components with different geometries can be fabricated in the same batch [28–34]. Accordingly, LPBF has attracted growing attention in recent years for producing numerous types of metallic components that mainly comprise aluminium [29,30,35–38], titanium [39–43], stainless steels [44–47], superalloys [48,49], nickel [50,51], and cobalt-chromium alloys [52–54].

Some frequently observed challenges exist despite the several advantages of LPBF processing. First, the LPBF process is characterized by rapid solidification and high-temperature gradients within countless molten pools that can go up to 105–108 K/s, resulting in thermal stresses that lead to segregation and non-equilibrium phase formation [55,56]. For instance, as-built Ti–6Al–4V exhibits an acicular/lamellar α' hcp phase (martensitic) in its microstructure owing to the high cooling rates in LPBF [31,41,57–59]. Additionally, extraordinary metastable cellular structures have been detected frequently in steels and Al and high-entropy alloys (HEA) [60,61]. Such structures are thought to be caused by dislocation alignment or chemical segregation that enhances the mechanical properties of fabricated parts [39,40]. Consequently, as-built Ti–6Al–4V fabricated parts are likely to have excellent tensile properties but poor ductility [62]. Second, microstructural imperfections such as balling and porosity can be induced by the LPBF process. These microstructural imperfections originate from process-induced defects caused by initial powder contamination, local voids, or evaporation after deposition of the powder layer [59,62–65]. Such defects likely result in high internal stresses and eventually cause failure, particularly under fatigue loading [62,65]. Finally, the stability, configuration, and behaviour of the molten pool have substantial impacts on the surface roughness, porosity, and roughness formed by the layer-wise build-up (staircase effect) [31]. These limitations cannot be avoided completely; however, much work has been done to control the physical phenomena related to these defects to produce full-density samples. In this review, the research status and advancements in LPBF processing are comprehensively discussed to demonstrate the potential of LPBF in achieving high manufacturing efficiency and forming accuracy to meet the demands of metallic components in numerous industries, including medicine, automotive, biotechnology, energy, communications, and optics. First, laser processing parameters in LPBF, which can induce residual stresses, are

discussed. Second, a review of recent research on common LPBF defects, including the types, formation mechanisms, and how these defects influence the microstructural and mechanical properties of as-built components, is provided. Suppression strategies for these defects are presented. Third, the microstructural evolution of the LPBF process, that is, phase transformation, precipitation, and grain growth that are influenced by laser processing parameters and post-treatment to minimise and homogenise the generated residual stress are systematically explained and presented, highlighting the unique microstructural features obtained by LPBF.

Fourth, recent studies on materials used in LPBF processing are reviewed. Because the application of the LPBF process is limited to mesoscale and macroscale fabrication, a short review is given to the recently developed micro-LPBF for 3D microfeature fabrication of various metals. Finally, the findings are summarized, and future research directions that solve the existing challenges of LPBF are suggested.

2. LPBF process parameters

The main challenge of using metallic powders in LPBF is determining the proper process window to fabricate 3D components with exemplary microstructural and mechanical properties. Accordingly, many studies have focused on optimising the LPBF process parameters. These studies can be roughly divided into three main categories: laser-related, powder-related, and powder-bed-related properties, as shown in Fig. 1.

2.1. Powder-related parameters

Powder-related parameters, including chemical composition, surface morphology, and particle size distribution, are considered invariants in the production environment [66]. The chemical composition of the powder significantly affects the LPBF process and its process parameters. The as-fabricated LPBF parts from the Hastelloy X and IN738LC compositions showed that minor alloying additives such as Mn and Si have a substantial impact on the crack sensitivity during the LPBF process [67,68]. Averyanova et al. [69] investigated the 17-4 PH martensitic stainless steel with slight variations in the chemical compositions of different powder batches. The findings revealed that the densification behaviour is not significantly affected by minor changes in the chemical composition. However, the chemical composition of the powder influences its microstructure and, consequently, its mechanical properties. Kempen et al. [70] reported that increasing the Si content enhances the processability (process parameters) of the AlSi10Mg alloy because Si improves the laser energy absorption of the powder bed. The final quality and density of the LPBF-processed parts are strongly influenced by the morphology of the starting powder.

Powder morphology can be used to determine the degree to which the powder particles are tightly packed during the deposition of a new powder layer on top of a solid layer. Thus, morphology plays a critical role in determining the layer thickness and surface roughness of the fabricated SLM part. Powder morphology can define the size and shape of powder

particles. Morphology depends strongly on the production methods, such as gas-atomization powder processing and mechanical alloying/mechanical milling. Chang et al. [71] studied the impact of particle size on the microstructural features, constitutional phases, and mechanical properties of LPBF-processed parts. The results have demonstrated that decreasing the powder particle size enhances the microstructural homogeneity of the part. Moreover, the fine particles also reduce the coefficient of friction (COF) and wear rate by 19% and 66%, respectively, from that of the LPBF part processed with coarse particles.

Niu et al. [72] demonstrated that spherical particles enhance the powder flowability and quality of built parts. Gas-atomization powder processing produces spherical particles [73], whereas mechanical alloying/mechanical milling produces irregularly shaped particles [74]. This, in turn, causes less compaction, resulting in increased porosity in the LPBF-fabricated components [75]. Moreover, gas-atomised powders promote uniform chemical composition [76]. Li et al. [77] reported that gas-atomised powder produces a denser structure than water-atomized powder because of the lower oxygen concentration and greater packing density of the gas-atomised powder. As seen in Fig. 2, the high-sphericity powder can be easily spread over the substrate using a scraper.

Gu et al. [78] have demonstrated that differences in particle size distribution result in differences in the relative density and thermal conductivity of the samples. Particle distribution and size can greatly influence powder viscosity, which can affect the transportability and dispersibility of the powder. In the case of large particle-sized powder, the inter-distances between the powder particles are not small. Consequently, a less dense structure is promoted. However, when the particle size is too small, agglomeration and adhesion of the powder are more likely to occur, and consequently, the powder fluidity declines. Thus, an effective forming effect can be achieved by combining coarse and fine powder particles during the LPBF process. The particle size ratio and particle sphericity must be optimised to guarantee the uniformity of the powder distribution and stability of the powder-dispensing system [79]. Baitimerov et al. [80] have investigated an AlSi12 alloy by employing three different powder batches that varied in morphology, particle size, and distribution. The results have demonstrated that the powder batch with particles has a near-spherical morphology. The high volume of fine particles shows poor flowability, which is ascribed to the formation of agglomerates, which in turn results in LPBF-built parts with high porosity levels.

2.2. Laser-related parameters

Laser-related properties, including laser system properties, such as the laser type (pulsed or continuous wave (CW)), spot size, and laser power, and scanning parameters, which include the scanning strategy, scanning speed, and hatch spacing, significantly affect the characteristics of the built component [89]. The properties and quality of LPBF-fabricated components are significantly affected by the parameters mentioned above, which in turn can be used to determine the energy density or energy input. The term "energy density" refers to the level of energy absorbed by a volumetric unit of

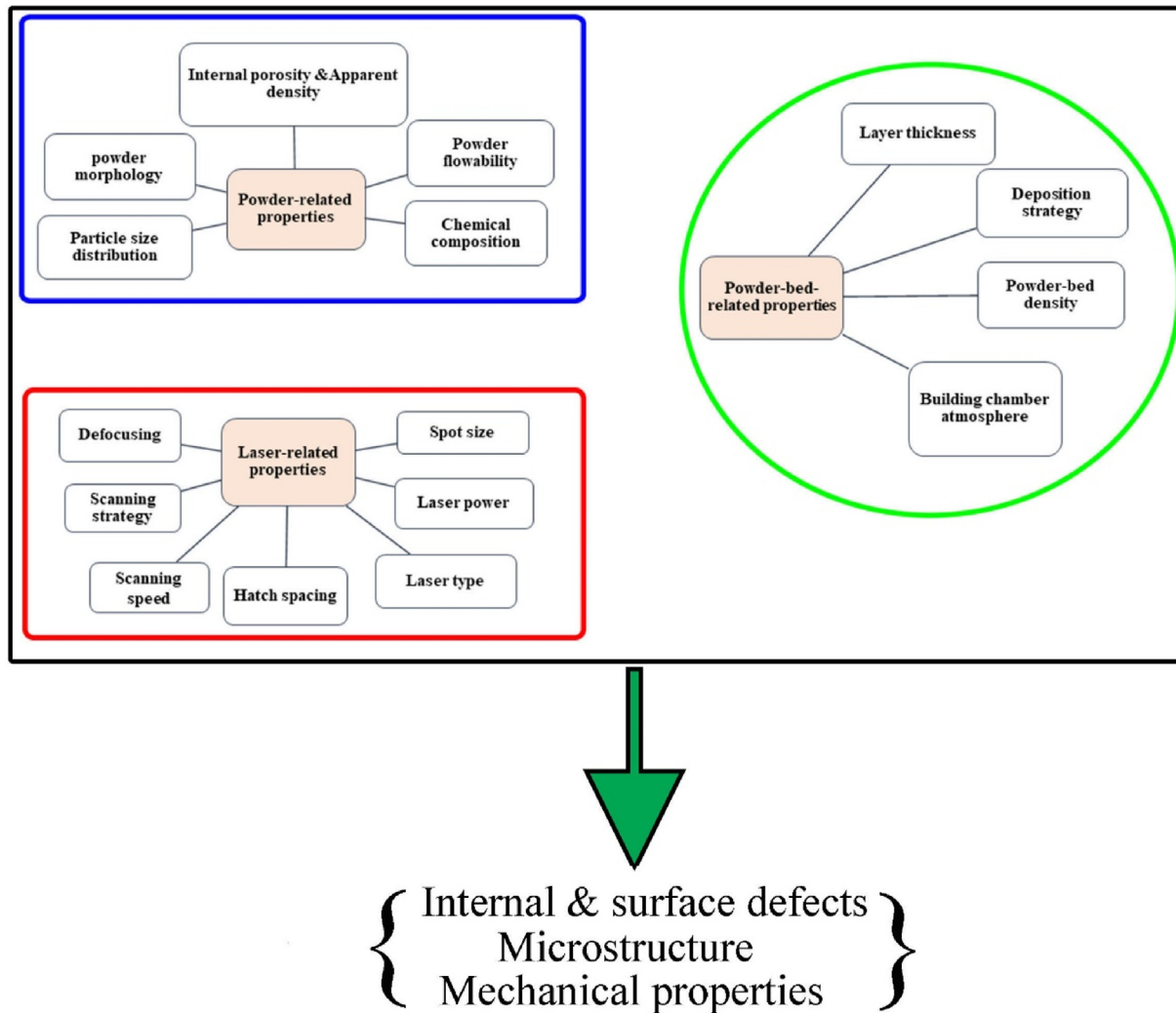


Fig. 1 – Schematic showing the three main processing parameters involved in LPBF: powder-related, laser-related, and powder-bed-related properties.

powder during the melting step. This can be estimated as follows:

$$E = \frac{P}{vht} \quad (1)$$

where P , v , h , and t represent the laser power (W), scanning speed (mm/s), scanning space or hatch spacing (μm), and powder-layer thickness (μm), respectively. Thus, a low scanning speed and/or high laser power, layer thickness, or hatch spacing increase the energy density of the laser, which is essential for heating and melting the powder. The increase in the energy density of the laser can increase the density of the fabricated parts, even under different atmospheres (argon, nitrogen, or helium), as shown in Fig. 3(a). However, there is a threshold energy level, 30 J/mm^3 for the Al–12Si components, below which the energy is insufficient to melt the powder entirely, consequently causing low density and higher porosity [90], as shown in Fig. 3(a). However, if it exceeds this level, it can be harmful to the surface finish because it can induce a balling formation [91]. Buchbinder et al. [92] have

fabricated AlSi10Mg parts using LPBF under varying laser power, as shown in Fig. 3(b). The findings have demonstrated that high-power lasers can achieve fast build rates and high densities. Prashanth et al. [93] have revealed that the strength properties of LPBF Al–12Si fabricated parts gradually decrease with decreasing laser power. Therefore, the energy provided to the powder bed decreases, resulting in inappropriate melting of the metal powder and solidification defects, which in turn results in poor mechanical properties.

Residual stresses and deformation are critical challenges for LPBF-built parts, limiting the wide implementation of LPBF technology. During LPBF, a highly localised heat input can cause a substantial temperature gradient, which, in turn, can lead to stress generation inside the fabricated parts. Furthermore, plastic deformation is induced when the thermal stress exceeds the material yield strength. The accumulation of stress and deformation during repeated thermal cycles produced by the layer addition process can lead to potential failure and severe distortion of the LPBF components, that is, cracking and layer delamination. These deformations and

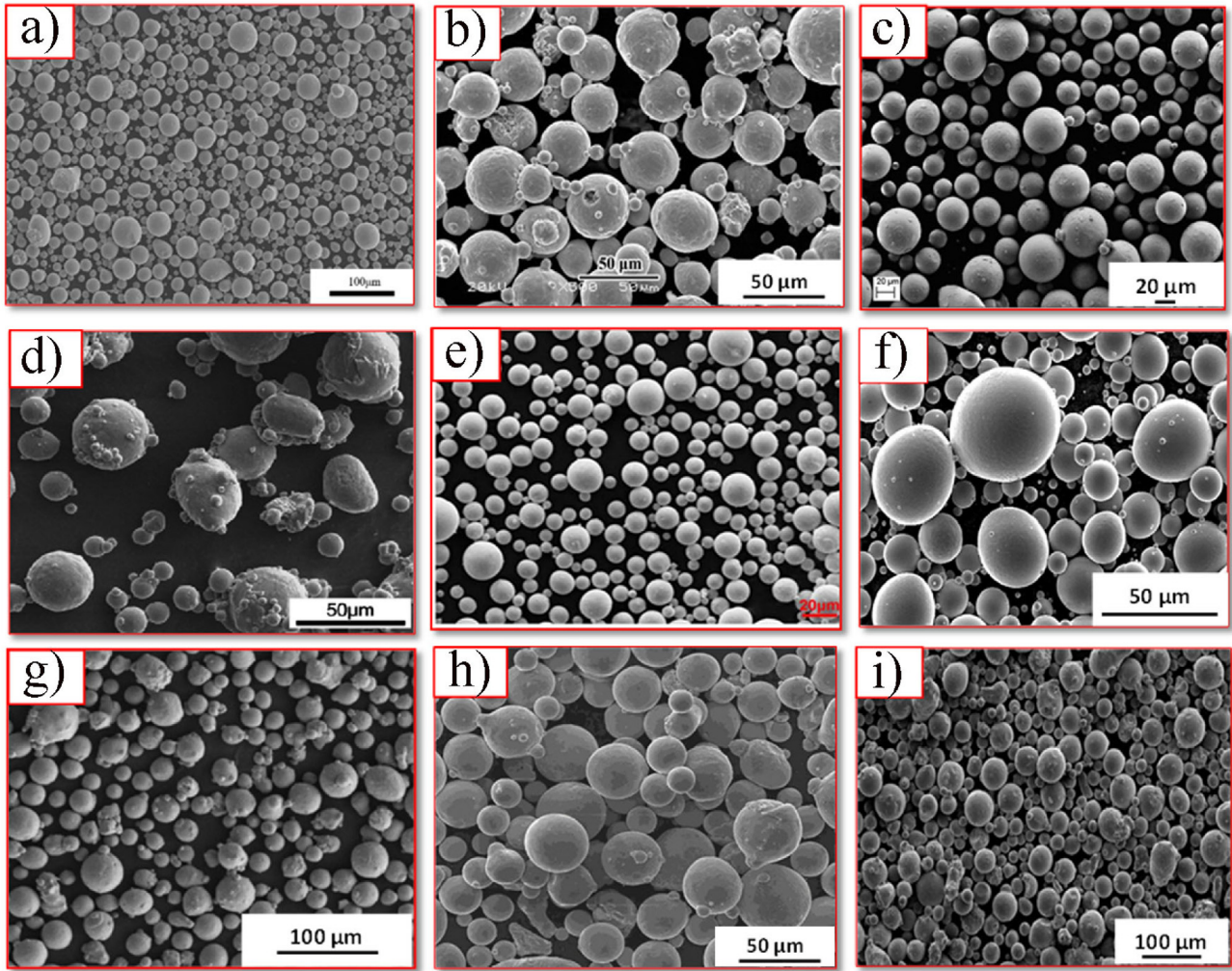


Fig. 2 – Metal powders: (a) 316L stainless steel [81], (b) 24CrNiMo alloy [82], (c) Ti–6Al–4V [83], (d) Al–12Si [84], (e) pure tungsten powder [85], (f) NiTi, (g) AlSi10Mg [86], (h) Inconel 718 superalloy [87], and (i) Hastelloy X alloy [88].

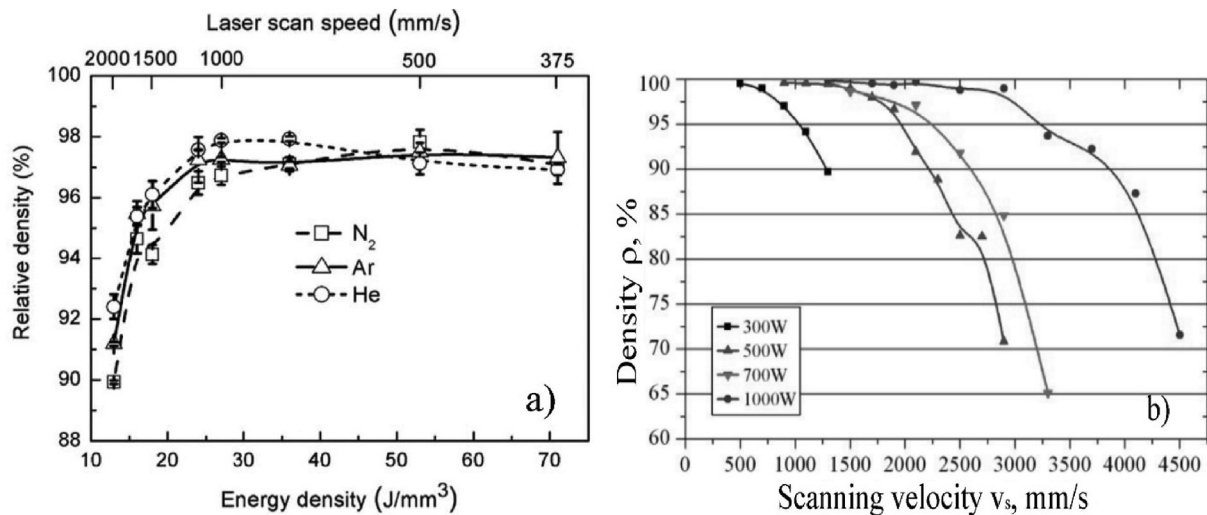


Fig. 3 – (a) Relative density in terms of incident laser energy for LPBF-processed Al–12Si components. An increasing trend of the relative density with laser energy density was observed. Below $E = 30 J/mm^3$, the energy required to completely melt the powder would be insufficient; the relative density plateaus above this value [84]. (b) Density of an AlSi10Mg12Si as a function of the scanning velocity at varying laser powers (0.15 mm scan spacing and 50 μm layer thickness) [92].

residual stresses are significantly influenced by the scanning strategy owing to variations in the local heat distribution [94]. Fig. 4 shows the most common scanning strategies used in LPBF. The scanning strategy in LPBF can be defined as the spatial moving pattern of the laser beam. For a single-layer scan, the pattern can differ according to the scanning direction [95], scanning sequence [96,97], scanning vector length [98,99], scanning vector rotation angle [100–105], scanning time [106], and hatch space [107].

Kruth et al. [108] have studied the influence of the scanning vector length and scanning pattern on residual stresses. They have found that island scanning decreases the residual stress with the greatest decrease realised after the scanning path on each individual island is inclined by 45° from the x-axis.

Furthermore, it is possible to reduce the generated stresses by reducing the scan vector length. Kruth et al. [109] have also studied the influence of different scanning patterns on deformation in an experimental setting. They showed that, unlike unidirectional scanning, sector-wise scanning, such as island scanning, can reduce directional distortion by revealing minor curvatures along the vertical or horizontal (x or y) directions in the fabricated specimens.

Liu et al. [110] have reported that cross-directional raster scanning produces a more uniform residual stress distribution than single-direction raster scanning, even though the residual stress level is nearly the same in both cases. Lu et al. [98] have studied the effect of island size on the residual stress of as-built LPBF Inconel 718 components. The findings show

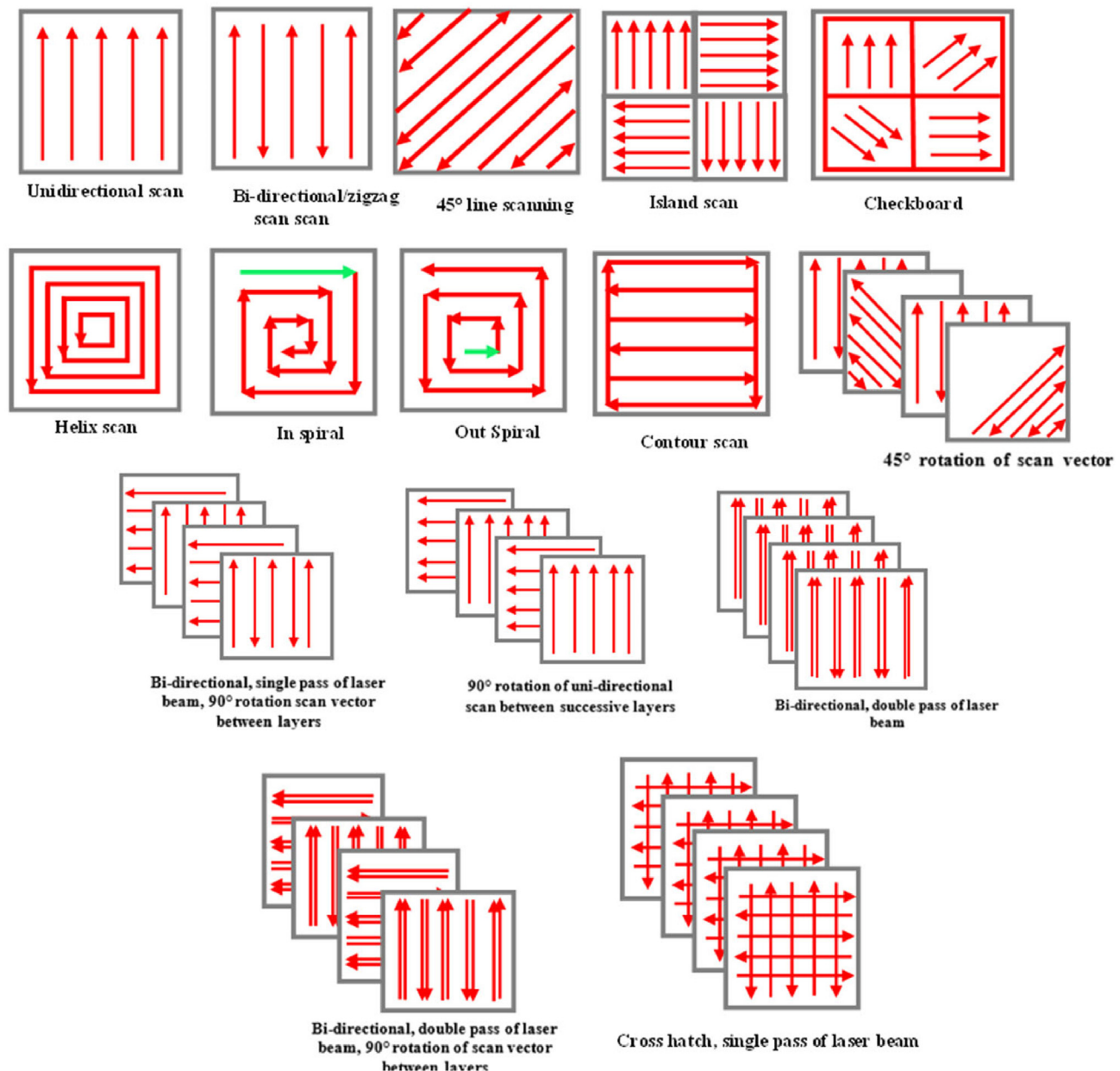


Fig. 4 – Schematic of different scanning strategies implemented in LPBF.

that samples with a 5×5 mm island size reveals less residual stress than samples with a 3×3 mm or 7×7 mm island size. Qian et al. [14] have conducted an experimental deformation analysis of LPBF-built parts using progressive raster scan and helix scanning approaches. The results indicate that the helix scanning approach is well suited for modelling complicated structures. Qian et al. [111] have performed a deformation analysis of processed LPBF-processed parts using progressive raster scan and helix scanning strategies. Their findings reveal that the helix scan strategy is effective for processing complex models. However, it is incompatible with the ring and cylinder models. The influence of the scanning pattern on substrate deformation has been investigated experimentally by Gao et al. [112]. The results show that the distortion of the substrate in the build direction is substantially larger than that in the other two orientations for all tested scanning patterns. Moreover, raster-scanned patterns may have a greater direct impact on the substrate deformation than contour-offset scanning patterns. Because raised edges are found in the SLM-produced part, Yasa et al. [113] have found that raster scanning, rather than unidirectional scanning, improves the flatness of the top surface by reducing the built-part edge height.

From the literature, it is clear that substantial research gaps still exist about the scanning strategy that has to be addressed to obtain the full benefits of additive manufacturing by coupling the processing parameters in LPBF.

3. Metallurgical defects

A promising future and wide-ranging application of the LPBF technology relies on manufacturing defect-free products. Despite significant efforts, numerous metallurgical defects in fabricated LPBF parts that can affect their mechanical properties remain unresolved. These defects include porosity, balling, cracking, and oxide inclusions.

3.1. Porosity

Porosity is inherent in most metals additively manufactured using powder-based AM techniques. Porosity affects the mechanical properties, fatigue strength, and fracture of AM products [114,115]. Particle size distribution significantly affects the porosity of the parts additively manufactured by LPBF, as reported by Nandwana et al. [116]. Their results show that the broad particle size distribution of Ti6Al4V induces low porosity. In contrast, a large particle size enhances porosity. Fig. 5 shows the internal and surface pores of AM materials fabricated by LPBF. Fig. 5(a) shows round pores with varying sizes in the AlSiMg alloy [117]. Similar round pores of various sizes are formed in 316 L stainless steel [115]. Rautio et al. [118] have reported surface pores with an elliptical morphology at the boundary between the two layers, as shown in Fig. 5(c). Furthermore, a near-round-shaped pore with microcracks promotes the outer surface, as shown in Fig. 5(d).

Porosity is generally small, typically less than $100 \mu\text{m}$, with an almost spherical shape [119], as shown in Fig. 6(a). These spherical porosities typically result from insufficient

melting (lack-of-fusion porosity) and entrapped gases in the molten pool owing to excessive energy input or instabilities in the process conditions. The former type can be ascribed to unstable process conditions resulting from a high scanning speed, low laser power, large hatch spacing, and layer thickness. The two main mechanisms can be classified according to the porosity position, with the former occurring outside the melt pool and the latter occurring inside. Fluid flow instability in the melt pool caused by multiple forces acting on the metallic/gaseous interface is a common characteristic in generating both pore types [120]. These pores are randomly dispersed in LPBF-processed parts and are challenging to eliminate. A more detailed illustration of the formation mechanisms of porosities can be found in [31,59,121–126].

3.2. Balling effects

Balling phenomenon can easily occur during LPBF, decreasing the quality of the produced surfaces. Balling is divided into two types: ellipsoidal balls with diameters of approximately $500 \mu\text{m}$ and spherical balls with diameters of approximately $10 \mu\text{m}$, as shown in Fig. 6(b) and (c), respectively. The former is ascribed to broken melt tracks caused by the poor wetting ability of the substrate or the previous layer. Both severe oxidation of the melt pool and low energy input can result in poor wetting characteristics [127].

The second type of balling is ascribed to laser-induced melt splashes due to the high capillary instability of the melt, which in turn induces a decrease in the liquid surface energy at short length scales [91]. Both types of balling effects can hinder the uniform deposition of fresh powder on top of the previous powder layer, resulting in additional defects such as porosities and cracks that can easily form between numerous discontinuous metallic balls, causing poor mechanical properties of the produced LPBF part.

Gu et al. [39] have shown that a low scan speed combined with a high laser energy density can induce microscopic balling defects and interlayer thermal microcracks. Aboulkhair et al. [121] have shown that the AlSi10Mg fabricated by LPBF exhibits balling effects when the scanning speed is higher than 500 or 750 mm/s . For a $5 \text{ wt } \% \text{ TiC/AlSi10Mg}$ composite, the threshold drops down to 350 mm/s , with other LPBF parameters remaining constant [128]. TiC/316L nanocomposites [129] and 316 L stainless steel [127,130] fabricated by LPBF exhibits a similar trend.

Balling phenomenon is influenced by the physical properties of the materials, that is, the volume of the nanoparticles has a significant impact on the threshold. Balling effect is detected at a scan speed of 150 mm/s in $7.5 \text{ wt } \% \text{ TiC/AlSi10Mg}$ nanocomposites and 350 mm/s in $5 \text{ wt } \% \text{ TiC/AlSi10Mg}$ [131]. Moreover, balling effect is influenced by the powder fabrication method, which in turn governs the shape and size of the starting powder. Gu et al. [132] have reported that LPBF-processed TiC/Ti nanocomposite parts prepared by mechanical mixing exhibit more severe balling than those processed by ball milling. The poor wettability of the directly mixed powder, which causes the melt to spheroidise rather than spread outwards on the underlying surface, contributes to melt pool instability.

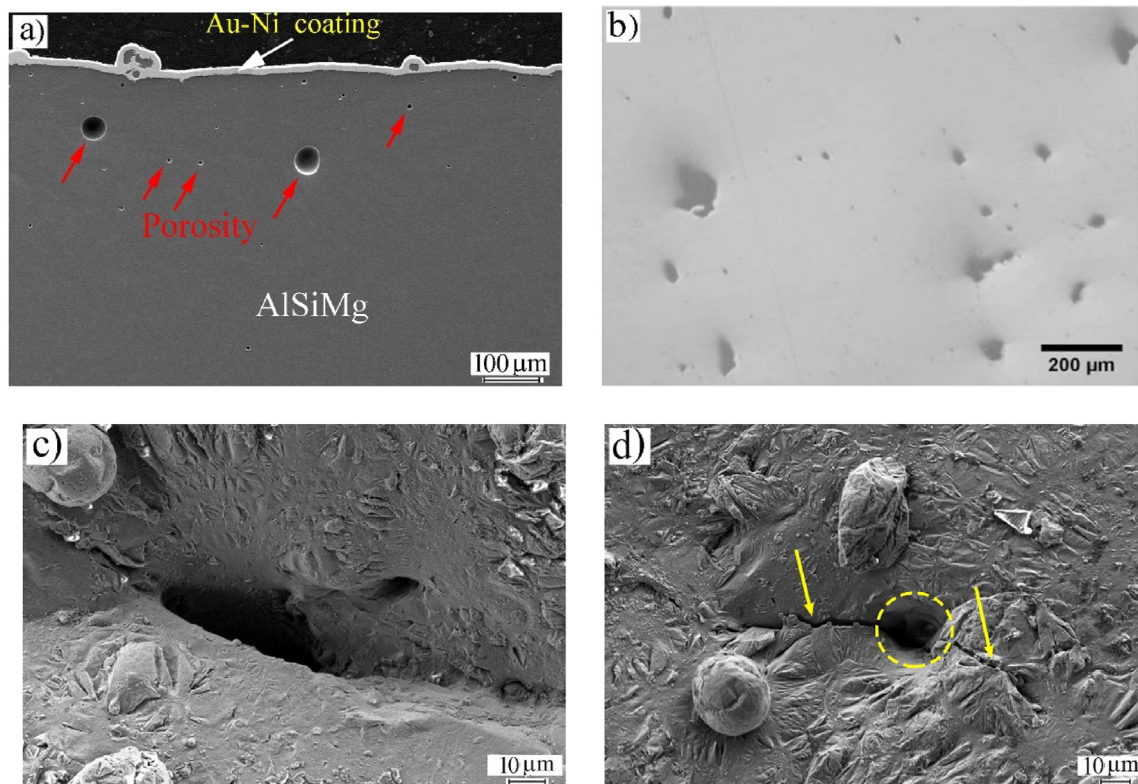


Fig. 5 – SEM images showing the internal and surface porosity in different materials during LPBF. Internal porosity in (a) AlSi10Mg [117] and (b) 316L stainless steel [115] and surface porosity in (c–d) AlSi10Mg alloy [118].

3.3. Incomplete fusion holes

A lack of energy input during LPBF results in incomplete fusion holes that are commonly referred to as lack-of-fusion (LOF) defects. LOF defects are caused by insufficient melting, which occurs when metal powders are not completely melted enough to deposit a fresh layer on top of the preceding layer with a sufficient overlap [132,135]. LOF defects can be classified into two types: (i) poor bonding defects resulting from an insufficient amount of molten metal during solidification, as illustrated in Fig. 6(d), and (ii) separated surfaces with unmelted powder particles, as shown in Fig. 6(e). The laser energy input significantly affects the molten pool width during LPBF. If the laser energy input is low, the molten pool width decreases, resulting in an insufficient overlap between the scan tracks. The insufficient overlap is responsible for forming unmelted metal powders between the scan tracks, making it difficult to remelt them entirely when depositing a fresh layer. Thus, incomplete fusion holes are formed and retained in the manufactured LPBF part. Moreover, if the laser energy input is insufficient to penetrate the molten pool deeply enough, poor interlayer bonding can cause LOF defects to form and disperse between the scan tracks and deposited layers [134–136].

3.4. Surface roughness

The regions where defects have been generated in the LPBF-fabricated parts, such as the LOF, unmelted powder, open

pores, material stacking, and balling effects, usually have a rough surface. These defects depend on the laser energy input and the size and properties of the powder particles. It has been reported that powders with large particle sizes are hard to melt and may result in increased surface roughness in fabricated parts [137]. The rough surface directly contributes to the poor flow of molten metal, resulting in interlayer defect formation. These interlayer defects can grow progressively and propagate upward in a continuous deposition process, eventually forming large multilayer defects [136]. Thus, poor surface roughness is a significant problem that requires post-processing.

3.5. Cracks

During LPBF, metal powders melt and rapidly solidify when exposed to a high laser energy input owing to the short laser–powder interaction time. Consequently, a large temperature gradient is generated in the LPBF part, leading to large residual thermal stresses. The high thermal gradient, along with the residual stress, causes the initiation and propagation of cracks in a fabricated part [60,138,139]. Two main types of cracks occur during LPBF: hot and cold cracks. The former can be generated at the end of the solidification stage when grains form continuous skeletons. Two phenomena contribute to the initiation of hot cracks: solid deformation and insufficient convection in the liquid region caused by a large temperature gradient during solidification [137]. Owing

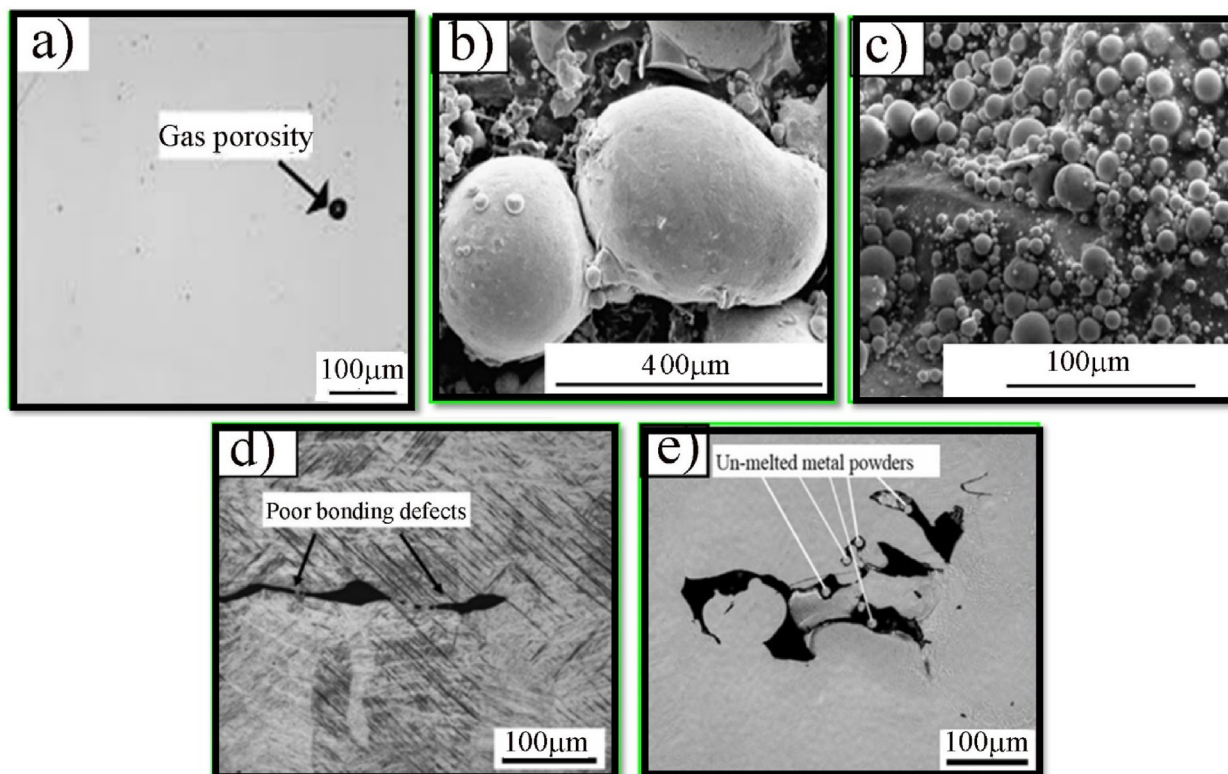


Fig. 6 – (a) Typical microstructure of the formed porosity in LPBF Alloy 718 [133], (b) typical microstructures of the first type of balling effect (characterized by highly coarsened balls; $P = 350$ W, $v = 0.04$ m/s and $d = 0.25$ mm) [91], (c) the second type of balling effect (characterised by a large amount of micrometre-sized ($10\ \mu\text{m}$) balls; $P = 400$ W, $v = 0.07$ m/s and $d = 0.25$ mm) [91], (d) microstructure of the first kind of LOF defect in the cross-sections of the LPBF fabricated Ti–6Al–4V parts (poor bonding defects) [134], and (e) the second kind of LOF defect (separated surface with unmelted powder particles) [134].

to the rapid heating of the upper surface by the laser beam and relatively slow heat conduction, a steep temperature gradient can be generated. The second type of crack that is more common in LPBF-built parts is induced by residual stresses. Two distinct mechanisms produce these residual stresses: the temperature gradient mechanism (TGM) and the cool-down phase of the molten top layers. TGM has been explained elsewhere [109]. The cool-down phase of the molten top layers includes shrinkage owing to thermal contraction. The top layer experiences tensile stresses, whereas the bottom layer experiences compressive stresses due to the underlying layers being prevented from deformation [140]. The formed residual stresses can lead to cracking and interlayer debonding; their levels depend on the material properties, particularly the elastic modulus.

3.6. Loss of alloying elements

LPBF is characterised by a high-temperature melt pool that, in turn, can cause substantial vaporisation of various volatile elements such as Mg, Li, Al, and Zn, with a high equilibrium vapour pressure. The equilibrium vapour pressures of various alloying elements as a function of temperature are shown in Fig. 7. Selective vaporisation of volatile elements during laser melting may result in changes in the composition and

properties of the fabricated parts [141]. Similar to the welding process, the vaporisation rate and melt-pool volume influence the change in metal composition. Despite the rate of vaporisation increasing with laser power, a change in composition is likely to occur at low laser powers owing to the small size of the melt pool, resulting in a high surface-to-volume ratio of the melting liquid.

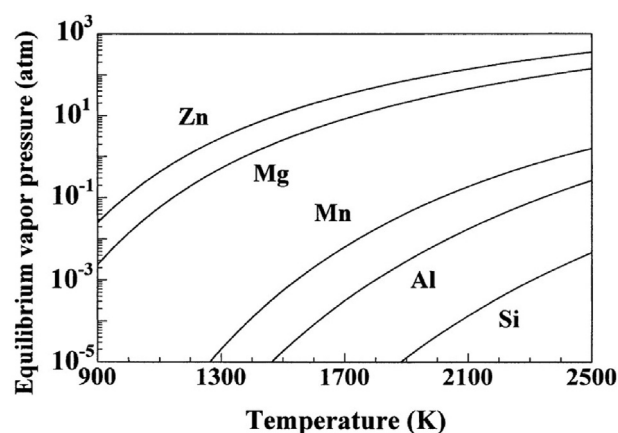


Fig. 7 – Equilibrium vapour pressure of metals at various temperatures [141].

3.7. Oxide inclusions

The oxide inclusions formed during LPBF entrap oxygen impurities and react with these impurities in the shielding gas owing to turbulent flow in the melt pool depression. Consequently, these oxide inclusions can harm the densification and quality of the LPBF parts. Thijs et al. [142] have found a large oxide inclusion comprising small TiN particles in 18Ni (300) parts of maraging steel. In their research, an oxide layer forms on top of each LPBF layer. When the previous layer melts, the oxide layer breaks and is dragged further.

Therefore, the final fabricated parts may contain accumulations of oxide materials that appear as large, irregularly shaped inclusions in the final fabrication process. Simonelli et al. [143] have investigated the potential impact of laser spatter on the processing of 316 L stainless steel, Ti–6Al–4V, and AlSi10Mg alloys. In their work, oxide films are found on the surfaces of 316 L stainless steel and AlSi10Mg spatters, but no oxidation is found on the Ti–6Al–4V powders.

Surface enrichment of the most volatile elements, such as Mg in AlSi10Mg, Mn, and Si in 316 L stainless steel, could lead to additional oxides. The oxidation films have two sides: wet and dry. The wet side acts as a nucleation site for intermetallics, whereas the dry side presents a potential nucleation site for porosity. As a result, most of the oxide films can be found at the pore boundaries [144].

4. Strategies for defect suppression

Defect suppression is challenging. Presently, the main strategy to suppress defect formation in LPBF is to implement optimal adjustment of process parameters that can eliminate the formation of porosities, voids, balling effects, cracks, and oxide inclusions. It has been reported that a narrow hatching space, small powder size, high laser power, and low scanning speed can suppress or reduce the aforementioned defects [80,145]. However, no processing parameters that can be used in LPBF processing and apply to most common materials have been established.

For instance, Kruth et al. [109] have demonstrated that high scan speeds together with high laser powers reduce balling effect during LPBF. The reduced balling effect is due to the quick resolidification of the melt pool behind the laser spot, when the molten track remained short. Conversely, previously published works [146,147] have found a reverse effect: at very low scanning speeds (1 mm/s) and low laser powers (10 W), no balling occurs. A potential reason for this behaviour is that the scan track broadens at these low speeds owing to heat conduction. According to Chen et al. [148], the balling effect can be suppressed by increasing the input energy density, which is accomplished by increasing the laser power and decreasing the scan speed. Regarding porosity suppression, Kasperovich et al. [149] have revealed that scanning speed is the most important parameter that affects the porosity fractions, as seen in Fig. 8(a); in contrast, hatch distance is shown to be the least sensitive (Fig. 8(b)). Laser power reveals the distribution of an optimal parameter range that allows for minimal defects (Fig. 8(c)). Fig. 8(d) shows that laser focus only results in an increased fraction of voids after being adjusted

far downward (positive focus range), underneath the powder bed surface. The literature has established that whereas fusion pores are correlated with high scanning speed and low laser power, keyhole pores are related to high laser power and low scanning speed. A low scan speed is desirable to achieve a good surface finish on manufactured parts because the extended time enables the molten pool to redistribute and produce a flat surface. Furthermore, Mumtaz and Hopkinson [150] have demonstrated that high laser powers could cause a reduction in the surface roughness because the recoil pressures created by higher energy flatten the molten pool and enhance the melt wettability, reducing the balling-induced surface roughness. Wang et al. [151] have found that the combined effect of both the scan power and scan speed determines the scanning track characteristics, such as track width, which is a critical parameter for improving the surface quality of LPBF-processed parts.

Wang et al. [151] and Yasa et al. [152] have proposed that re-exposing the top surface of the LPBF-processed components to the laser after the original scan could achieve a sound top surface quality because re-exposure facilitates a remelting effect. For the vertical-side surface, Mumtaz [150] has recommended fast scanning to reduce the thermal variation across the molten pool and control the scanning track stability to achieve optimal side surface roughness. Thus, new approaches for eliminating and/or controlling the defects formed within built-in components should be further developed.

5. Solidification behaviour of LPBF

Metallurgical theories in welding [153] can be readily employed to qualitatively describe the characteristics of the grain structure because the materials in both welding and additive manufacturing processing are exposed to a moving heat source with a high energy input [154]. The grain structure of a single laser track can indicate the mechanism of solidification during LPBF based on the solidification behaviour in laser welding, as shown in Fig. 9. Metal additive manufacturing processes can produce both columnar and equiaxed grains. Columnar grains are normally developed epitaxially from the substrate or the first deposited layers and towards the heat-source direction [155,156]. By contrast, equiaxed grains can exist together with columnar grains or between every two layers of columnar grains, forming a sandwich grain structure, as observed in Ref. [157]. Nucleation preferentially occurs at the fusion line during solidification owing to its low activation energy. The nuclei prefer to align with crystallographic orientations identical to the partially melted grains in the lower layer. Epitaxial nucleation is a term that describes the nucleation mechanism [153]. Following epitaxial nucleation, grain growth occurs across the fusion line and along the local temperature gradient direction, which is considered the local movement direction of the solidification front [154]. Columnar grains are formed as a result of this directional solidification, as depicted in Fig. 9(a) by the purple grains. The idealised solidification mode, known as a competitive mechanism, can transform from planar to cellular, columnar dendritic, and equiaxed dendritic across

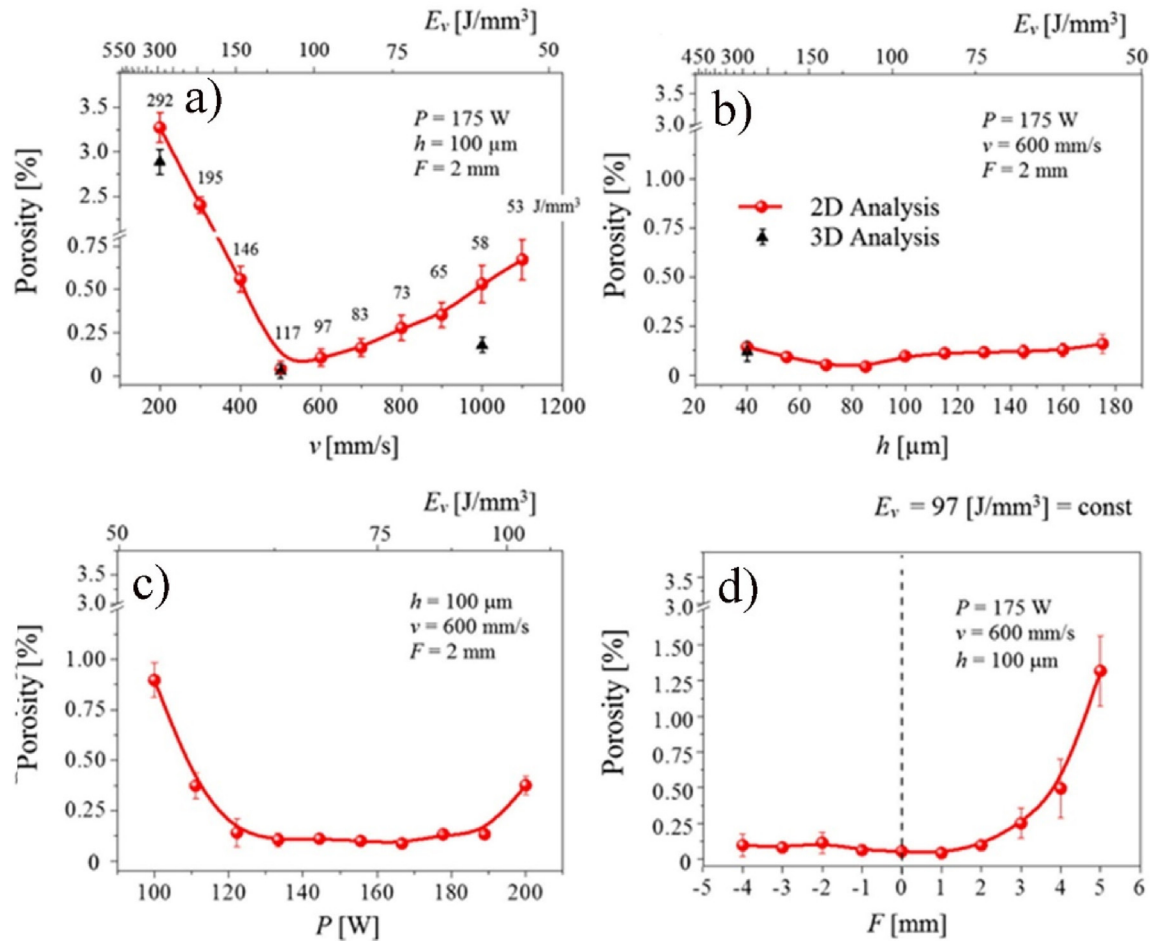


Fig. 8 – Effect of the LPBF process parameters on the total porosity in terms of (a) scan velocity, v , (b) hatch distance, h , (c) laser power, P , and (d) focus distance, F . The volume energy density (E_v) is shown on the top of each graph [149].

the scanning zone, as described in the classical solidification theory in welding metallurgy [153,158,159]. Equiaxed dendritic structure occurs because the G/R ratio (G : temperature gradient and R : solidification rate) decreases from the fusion line to the top of the molten pool, as shown in Fig. 9(c). Equiaxed grains can exist together with columnar grains or even prevent the growth of columnar grains, a phenomenon known as the columnar-to-equiaxed transition (CET), as shown in Fig. 9(a). This solidification mode, particularly the CET, can describe the solidification behaviour and microstructures of additive manufacturing alloys, such as the cellular-structured eutectics Al–Si [60,160] and B4C/Ti6Al4V [161], CoCrFeMnNi high-entropy [162] and Al_xCrCuFeNi₂ high-entropy alloys [163], the lamellar-structured eutectic Al–Al₂Cu in LPBF Al–33Cu alloy [164,165], and 304 and 316 L stainless steels [166].

Various studies have been conducted to develop analytical CET models that describe the grain structure in additive manufacturing [167,168] and forecast the vol % of equiaxed grains based on the local thermodynamic conditions of the solidification front [169,170]. However, the qualitative theories in welding and analytical CET models cannot completely and precisely predict the complicated grain structures and

orientations from the combined mechanisms in AM [154] owing to the random grain orientation of the grains formed in the preceding layer during layer-by-layer processing.

Cellular automata (CA), phase-field (PF), and Monte Carlo (MC) methods are the most widely used models to simulate the grain structure in additively manufactured metal alloys. Zhang et al. [171] and Rai et al. [172] have applied two-dimensional (2D) models to explore epitaxial nucleation and competitive growth mechanisms. The results suggest that the resulting columnar grain structure and large grain size regenerate, similar to the experimental results. Zinovieva et al. [173] have used a 3D-CA model to evaluate the grain structure evolution in multipass and multilayer simulations. The results show that under certain solidification conditions in LPBF, the development of coarse columnar grains with the most favourable growth direction misaligned with the build direction is enhanced. Shi et al. [174] have used a 3D-CA model to investigate CET through a single-track laser during the AM process under various laser beam shapes. Xuxiao and Wenda [154] have implemented a 3D-mesoscale CA model incorporated with a macroscale thermal model to simulate the 3D grain structure of LPBF-processed stainless steel 304. The results show that without any bulk nucleation (i.e. nuclei

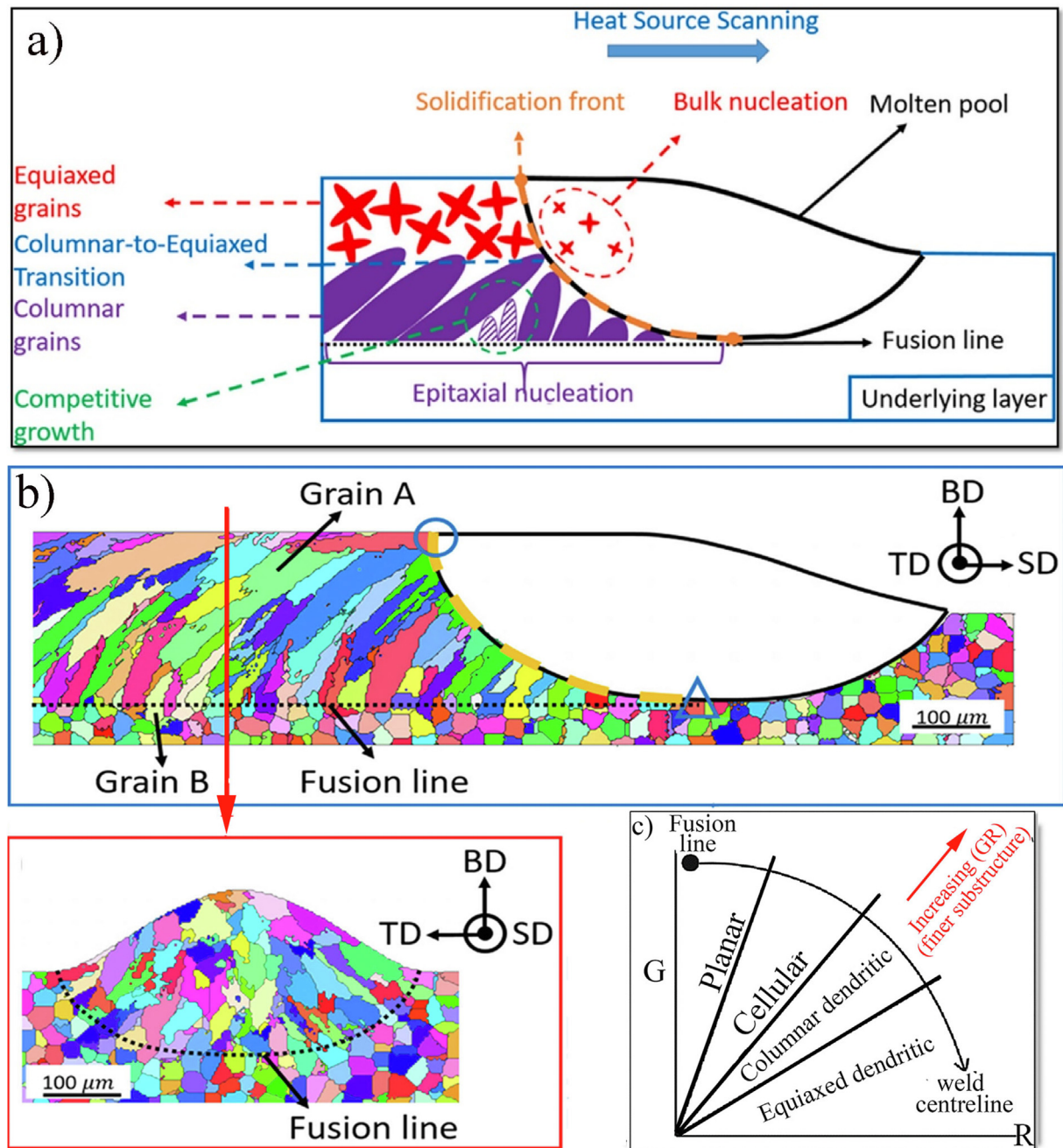


Fig. 9 – (a) Solidification behaviour of the center plane across scanning direction with possible nucleation and growth mechanisms, (b) Single-pass grain structure simulation during the building process without bulk nucleation [154], and (c) Schematic showing the influence of temperature gradient, G , and solidification rate, R , on the grain structure [182,183].

density is zero), the columnar grains dominate the grain structure and are likely to grow perpendicular to the moving solidification front, as indicated by the orange dashed line in Fig. 9(b). Besides, Fig. 9(b) demonstrates the competitive growth mechanism in which some grains, such as grain B are outgrown by more “favoured” grains such as grain A, whose $\langle 1\ 0\ 0 \rangle$ directions are better aligned with the local temperature gradient direction, i.e. the direction of the solidification front. The top zone of the molten pool may induce the formation of

equiaxed grains by increasing the nuclei density and/or decreasing critical undercooling.

Liu et al. [86] have demonstrated the validity of a simulated grain map of the cross-section along the building direction, as shown in Fig. 10(a). Their studies also demonstrate that the scanning speed and CET criterion of the LPBF-processed Al10SiMg alloy, which is defined by the G/V ratio, impact the grain structure variation. The CET transition occurs when the G/V ratio is less than a certain value, resulting in equiaxed

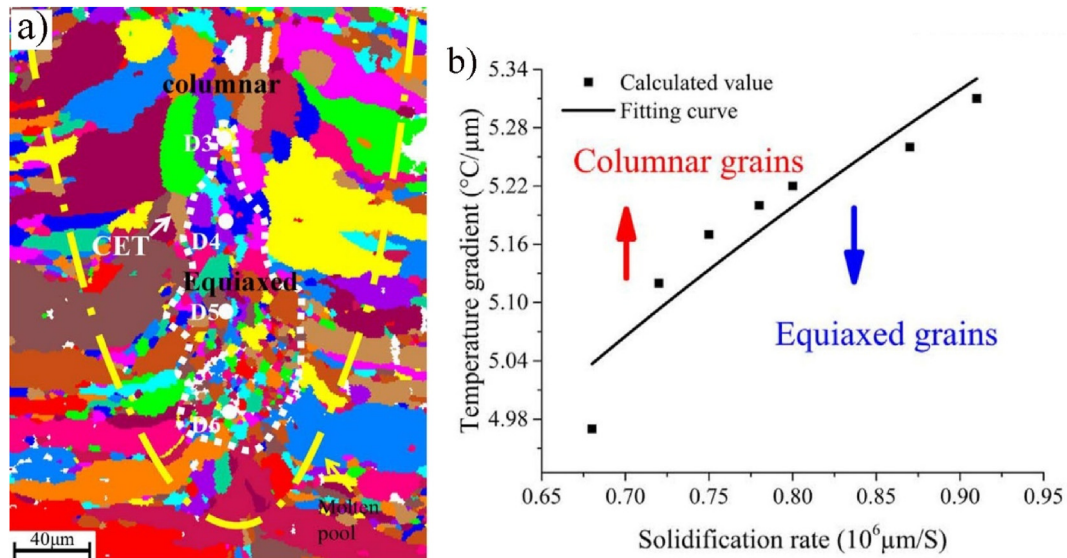


Fig. 10 – (a) Grain structure along the building direction and (b) the relation between temperature gradient and solidification rate [86].

grains, as shown in Fig. 10(b). The PF method, which is mostly employed in 2-D, is a highly effective tool for simulating microstructure evolution because it is based on thermodynamics and physical characteristics. Lu et al. [175] have developed a 2D-PF model to simulate and evaluate the effects of process parameters on the grain structure of the melt pool and morphologies in AM processing. Liu et al. [176] have established a temperature-dependent 2D-PF model to simulate grain growth during multilayer single-track scanning. Liu et al. [177] have investigated the mechanism of CET during a single-track process by combining a nucleation model with a 3D-PF model. However, few 3D-PF simulations of grain evolution in AM processes exist, and the results are limited to a single track. O'Toole et al. [178] have implemented a non-equilibrium CALPHAD-integrated PF model to capture the chemical kinetics of the phase transition during the fast solidification of an Al/Si alloy under AM conditions.

MC is also a model based on minimum grain boundary interface energy. The MC method provides higher calculation efficiency than the CA method but with some drawbacks [179]. The MC technique cannot accurately determine the crystal orientation. Moreover, the MC model requires an external model to provide information regarding the temperature field and time step, complicating the calculation parameters.

Studies in Refs. [106,180],[181] [182] have implemented the MC method to predict the microstructural evolution of metal alloys in AM. Jiang et al. [179] have provided a comprehensive review of the three models mentioned above and their applications in the numerical simulation of the AM process.

6. Grain growth during LPBF

6.1. Grain growth mechanism

Grain growth occurring during LPBF is a localised solidification process, similar to traditional welding, which begins with

epitaxial growth and progresses to competitive growth towards the centre of the melt pool [182]. Once the laser beam leaves the melt zone, the partially melted grains from the substrate or preceding layer solidify via epitaxial growth. The grain morphology and substructure, as previously mentioned in Fig. 9(c), are significantly influenced by the G/R ratio, consistent with solidification theory. During solidification, the grain growth orientation aligns with the direction of the temperature gradient. The competitive growth of different grains causes grains with similar orientations to grow fast, whereas grains with different orientations grow slowly [184]. Furthermore, when neighbouring grains impinge on each other, the horizontal grain growth is hindered, and the high rate of elongated grain growth reduces the possibility of nucleation occurring in the liquid ahead of the columnar zone [185]. During the solidification process, a unidirectional columnar grain structure (a microstructure typical in LPBF-processed components) is formed because of the competitive growth mechanism. However, by adjusting the laser parameters and/or adding impurities such as nanomaterials and using dynamic techniques such as arc oscillation, the nucleation process can be enhanced, resulting in equiaxed dendritic growth and fine grain structures [186–189].

The laser parameters, including laser power, hatch spacing, scanning speed, and layer thickness, determine the temperature gradient, cooling rate, and local heat flow direction; these parameters significantly impact the grain morphology, size, and orientation. Savalani et al. [190] have investigated the effect of the scanning speed and laser power density on the grain size of LPBF magnesium. The results show that increasing the laser power leads to an increase in grain size owing to the high energy-induced coarsening effect, whereas increasing the scanning speed results in a decrease in grain size owing to a short exposure time and fast solidification process. Similar results are obtained for stainless steel 316 L [102] and cobalt alloys [191]. Olsen et al. [182] have investigated the influence of the hatching spacing on

the grain morphology of LPBF-processed CoCrMo alloys. Owing to constitutional undercooling during LPBF, small equiaxed grains are frequently observed in the centre of the melt tracks. The new melt track remelts the centre zone as the hatch spacing decreases, triggering significant epitaxial grain growth over the whole region. Scanning speed, which affects the grain morphology throughout the LPBF process, is another crucial parameter to consider. Gu et al. [85] have studied the grain morphology of pure tungsten during LPBF using three scanning strategies: chessboard, zigzag, and remelting. In the chessboard scanning strategy, the microstructure of the as-built LPBF tungsten exhibits equiaxial columnar grains split into multiple areas by cracks from the top view, as shown in Fig. 11 (a). From the front view in Fig. 11(d), the columnar grains are aligned in the building direction. The formation of columnar grains is typically affected by the direction in which the temperature decreases, following classical solidification theory [192]. In addition, cracks and residual pores are detected in both top and front views. Fig. 11(b) shows the zigzag scanning strategy, which results in equiaxial grains with a consistent orientation throughout the scanning direction, as shown in the top view. The front-view microstructures exhibit slender columnar grains, as shown in Fig. 10(e). However, in both the top and front views, the cracks are greatly minimised. In the laser remelting scanning strategy, the microstructures of both the top and front views are coarse, as shown in Fig. 11(c) and (f). However, the cracks are significantly minimised. Thijs et al. [31] have investigated the effect of scanning strategies such as unidirectional, zigzag, and crosshatching on the grain morphology of the Ti–6Al–4V alloy during LPBF. The results show that all strategies can generate grain growth directions. Wang et al. [96] have introduced a comprehensive review focusing on the impact of scanning strategies on the microstructure during LPBF.

6.2. Effects of additives on grain growth

Adding additives, that is, second-phase nanoparticles, to the powder bed can drastically change the grain structure formation during LPBF. Xi et al. [193] have studied the influence of adding a ceramic mixture of TiB₂ and γ -Al₂O₃ to AlSi40 alloys manufactured via LPBF. By adding these nanoparticles, the primary Si phases become refined because they can act as heterogeneous nucleation sites for precipitating primary Si phases to prevent their coarsening. Li et al. [187] have investigated the grain structure of LPBF -AlSi10Mg alloys reinforced with TiB₂ nanoparticles and have observed that the addition of TiB₂ nanoparticles reduces the average grain size of AlSi10Mg from ~10 μ m–2 μ m. Similar results are obtained when fabricating Ni/Al₂O₃ [194], Al–Si–Mg–Ti/TiC [195], AlSi10Mg/TiN [196], and IN718/TiC nanocomposites [197] using LPBF.

In addition, some studies have found that nanoparticles do not always result in grain refinement. For instance, Streubel et al. [198] have observed that adding nano-Y₂O₃ has no effect on the grain morphology of an as-built iron-chromium component because of the poor wettability and lattice mismatch between the metal matrix and nanoparticles. Therefore, investigating the influence of nanoparticles on grain structure during LPBF is essential.

7. Phase transformation during the LPBF

As mentioned before, LPBF is characterised by high-temperature gradients and a rapid solidification rate, resulting in thermal stress build-up that causes segregation and/or non-equilibrium phase generation [182]. In LPBF of steels, steel powders such as 304 L and 316 L, can exhibit complete transformation to an austenitic microstructure as reported in Refs. [199,200] owing to the rapid melting and cooling rates. However, Zhu et al. [201] have observed that the as-built 304 L stainless steel (SS) comprised two phases: austenite and δ ferrite. The formation of δ -ferrite is also detected in 304 L SS [202,203] manufactured by directed energy deposition. The formation of δ -ferrite can be explained by the rapid cooling rate that prevents the complete transformation from δ -ferrite to γ -austenite [202–204]. As well, for 316 L stainless steel, Yadollahi et al. [194] have observed a small volume fraction of δ ferrite (5–10 vol %) regularly dispersed through the austenite grains.

Lebrun et al. [205] have observed 36 vol % austenite in LPBF-built 17-4 PH using different laser processing parameters. The results show that different amounts of metastable austenite are detected in martensitic steels after LPBF [206,207]. Sun et al. [208] have processed a Ti–6Al–4V alloy using LPBF under different processing heights. The results show that the Ti–6Al–4V alloy undergoes a regular transformation as the height of the sample changes. The microstructure mostly contains martensite α' because of the high temperature gradient. As the height of the LPBF -part increases, the lower parts are subjected to frequent thermal cycles, and the accumulation of the remelting heat results in microstructure coarsening. Thus, as the height of the specimen increases, the microstructure evolves according to the following rule: coarse lamellar α' —fine articular α' —ultrafine Z-zigzag α' . As shown in Fig. 12, the development and evolution of the minor phase, which consists of secondary α' , ternary α' and quartic α' phases, are highly correlated with the temperature cycle during LPBF. In addition, an open-cell porous Ti–6Al–4V structure has been produced via LPBF using commercial Ti–6Al–4V powders, in which the structure contains a hexagonal closed packed (HCP) α -Ti phase and a body-centred cubic (BCC) β -Ti phase [209]. The results show that after laser processing, the β phase transforms into a martensitic α' phase. Similar findings are also reported in Ref. [210].

8. Precipitation during LPBF

Precipitation is commonly detected in Ni, Al, and Mg alloys fabricated via LPBF. Similar to Ti alloys, LPBF precipitation may occur concurrently with phase transformation. The type and characteristics of precipitates that can greatly affect the mechanical properties of as-built LPBF parts are mostly dependent on the laser processing parameters and/or the powder chemical composition.

For Al/Mg alloys, Xiang et al. [211] have observed Al₃(Sc_xZr_{1-x}) precipitates in LPBF-fabricated Sc- or Zr-modified Al–Mg-based alloys. Precipitates are seldom observed in conventional casting. Ma et al. [212] have processed an Al–20Si–5Fe–3Cu–1Mg alloy via LPBF and have observed that

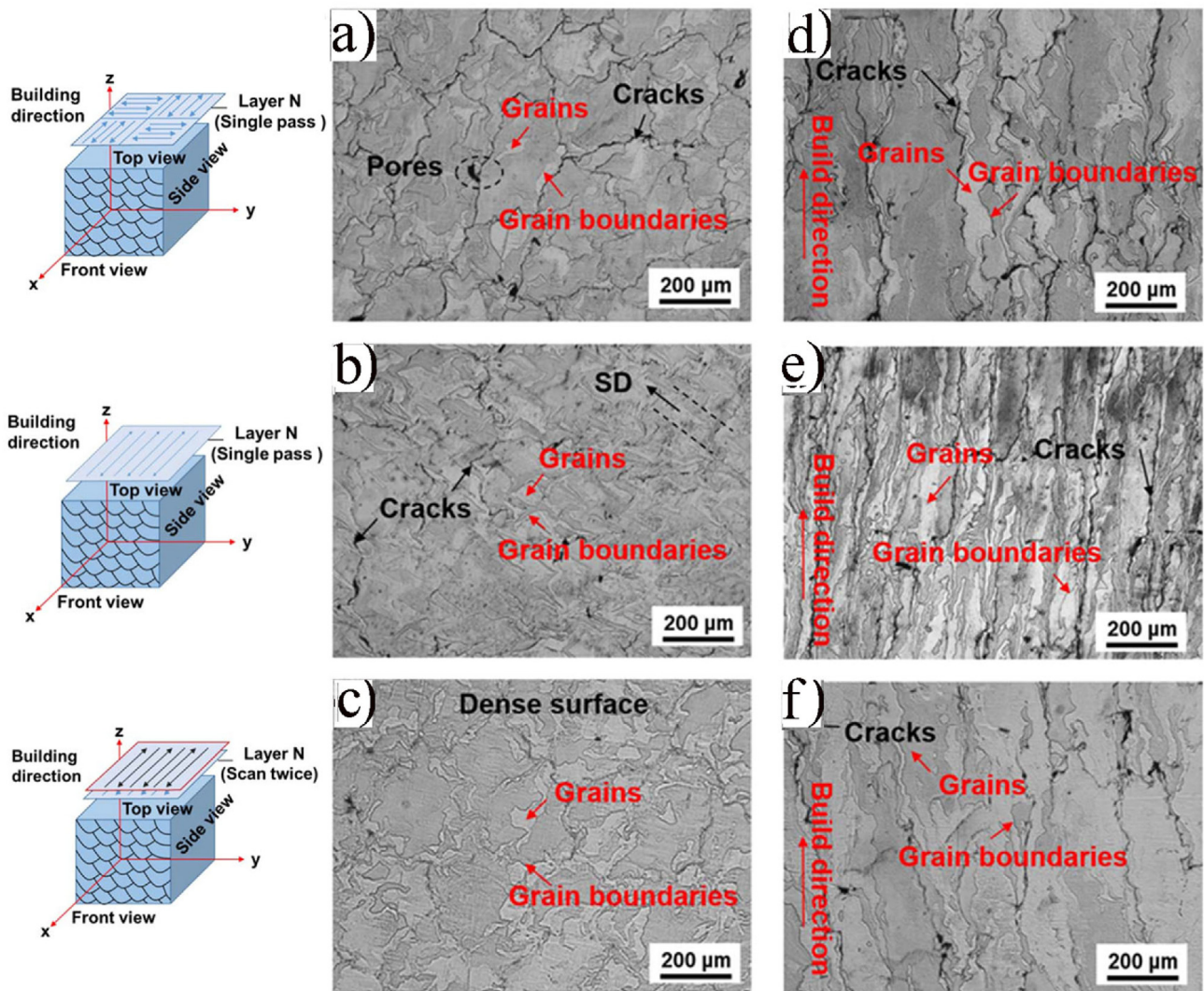


Fig. 11 – Microstructures of as-built LPBF tungsten when employing three different scanning strategies. (a) Top and (d) front views for the chessboard scanning strategy, (b) top and (e) front views for the zigzag scanning strategy, and (c) top and (f) front views for the remelting scanning strategy. SD signifies the scanning direction [85].

the as-built LPBF part only contains the δ -Al₄FeSi₂ phase, while the as-cast alloy generally comprises both β -Al₅FeSi and δ -Al₄FeSi₂.

For Ni-based superalloys, Idell et al. [213] have observed two potential precipitate phases formed in as-built nickel-based superalloys. These precipitates are the η -phase (Ni₆AlNb) and the unexpected δ -phase (Ni₃Nb). As the part is produced, the underlying layers undergo various heating and cooling cycles that cause considerable microsegregation of niobium, allowing for the formation of a detrimental δ -phase.

For titanium alloys, Qngsong et al. [214] have reported that the precipitation kinetics occurring in the LPBF-processed Ti–45Al–2Cr–5Nb alloys experience a $\beta \rightarrow \alpha_2$ (Ti₃Al) + γ (TiAl) transformation and disorder-order transformation ($\beta \rightarrow B2$). They have also observed that the α_2 phase decreases, whereas the γ and B2 contents increase when the laser scanning speed is increased. Lore [31] has studied the microstructural evolution of Ti–6Al–4V and found that

intermetallic phase Ti₃Al is precipitated during the process at high heat inputs.

Kuźnicka et al. [215] have observed that the as-built Ti–6Al–7Nb alloy produced by the LPBF has a layered microstructure with thin plates of α' martensite hardened by evenly distributed precipitates of the β -AlNbTi₂ phase. Precipitation of the phase β -AlNbTi₂ from martensite may be ascribed to the microsegregation of solute in β grains owing to their dendritic-cellular growth, resulting in improved tensile and compressive strengths of the LPBF-built components as a result of the precipitation-hardening effect.

In addition to the above-mentioned precipitates, the rapid cooling rate during LPBF can result in a supersaturated solid solution and inhibit precipitation, notably for Al–Si and Al–Sc alloys. Takashi et al. [216] have studied the microstructure of AlSi10Mg fabricated by LPBF and observed that the average Si concentration in α -Al in an LPBF sample is 2.2%, greatly surpassing the maximum solubility of Si (1.4 at %) in AlSi10Mg at

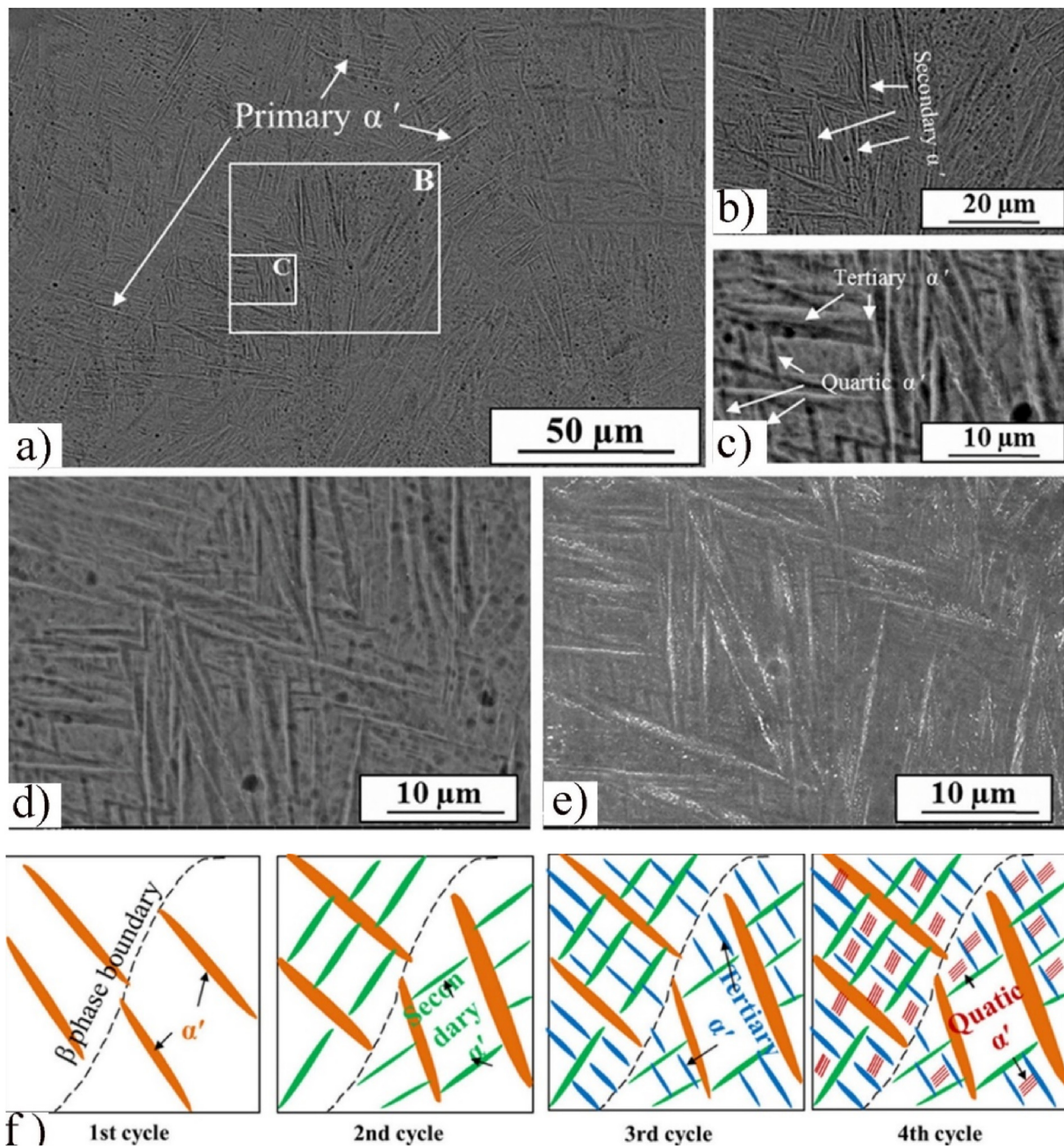


Fig. 12 – Classification and distribution of the martensite phase in Ti–6Al–4V alloy produced by the LPBF: (a) Scanning electron microscope image of the top surface, (b) and (c) enlarged view of the white box marked B and C in (a), respectively, (d) backscattered electron and (e) secondary electron images of top surface, and (f) schematic of the martensite transformation showing the effect of thermal cycles [208].

room temperature, i.e., the equilibrium state of AlSi10Mg. Dongdong et al. [217] have revealed the mechanisms of the precipitation distribution at various scanning speeds during the LPBF of Sc-modified Al–Mg alloys. The results showed that a small number of spherical nanoprecipitates, $\text{Al}_3(\text{Sc}, \text{Zr})$, are embedded at the bottom of the melt pool at a low scan speed (1800 mm/s). However, by increasing the scanning speed to 3000 mm/s, supersaturated solid solutions are formed, and no precipitates are detected due to the combined effects of varying Marangoni convection vector, an ultrashort lifetime of the liquid, and rapid cooling rate.

9. Effect of heat treatment

Heat treatment is considered as one of the most effective techniques for minimising and homogenising the generated residual stress. Wu et al. [208] have reported that after annealing, the residual stress of the as-built Ti–6Al–4V alloy decreases in both the scanning and normal directions by 59.8 and 72.3%, respectively. After solution aging, the residual stress declines in the same direction by 64.7 and 67.8%, respectively.

Post-treatment has been shown to significantly affect the microstructure and mechanical properties of as-built LPBF components. Gu et al. [218] have applied three different heat treatments to achieve the desired mechanical properties of as-built WC/Inconel 718 composites: homogenisation + double aging treatment (HA), solution + double aging treatment (SA), and homogenisation + solution + double aging treatment (HSA). The findings demonstrate that precipitates of approximately 200 nm in diameter are found in the microstructure, and the grain size increases to 34.13 μm with the reduction of high-angle grain boundaries after treatment. The heat-treated composites exhibit improved microhardness and tensile strength but reduced ductility. Homogenisation + double aging (HA) samples achieve superior mechanical properties among all samples, as shown in Fig. 13.

The improved mechanical properties can be ascribed to the combined effect of reduced molten pool boundaries (MPBs), residual stress, uniform precipitates, and low-angle grain boundaries. Bai et al. [219] have shown the regularity of microstructure evolution, microhardness, tensile performance, and impact toughness of maraging steel processed by LPBF after different heat treatments. Prashanth et al. [30] have shown that by applying an appropriate annealing treatment, the mechanical behaviour of Al–12Si can be adjusted within a wide range of strength and ductility. Moreover, post-heat treatment of the as-built ultra-high-carbon steel at temperatures between 700 and 1200 $^{\circ}\text{C}$ changes the microstructure

of the samples from austenite and martensite to fine and coarse pearlite, respectively [220]. Thus, post-processing heat treatment should be applied to enhance the mechanical properties of LPBF-fabricated parts [221–224].

10. LPBF materials

LPBF process can be applied to a wide variety of commercial metallic powder materials. Nevertheless, there are two important requirements for metals that may be treated by LPBF: good weldability to prevent cracks during solidification and raw material availability in the form of spherical powders that are tens of micrometres in diameter to achieve adequate packing density and powder deposition homogeneity. Numerous review articles have provided an overview of materials treated using the LPBF:

1. High-entropy alloys (HEAs) [225]
2. Ti-based alloys for medical applications [39,226–228]
3. Ni-based superalloys for jet engines and land-based gas turbines [229,230]
4. Al-based alloys [158,231,232]
5. Bulk metallic glasses [233,234]
6. Inconel and nickel-based alloys [235,236]
7. Ceramics [235,237–239]
8. Co–Cr alloys [240,241]
9. Magnesium-based alloys [242,243]

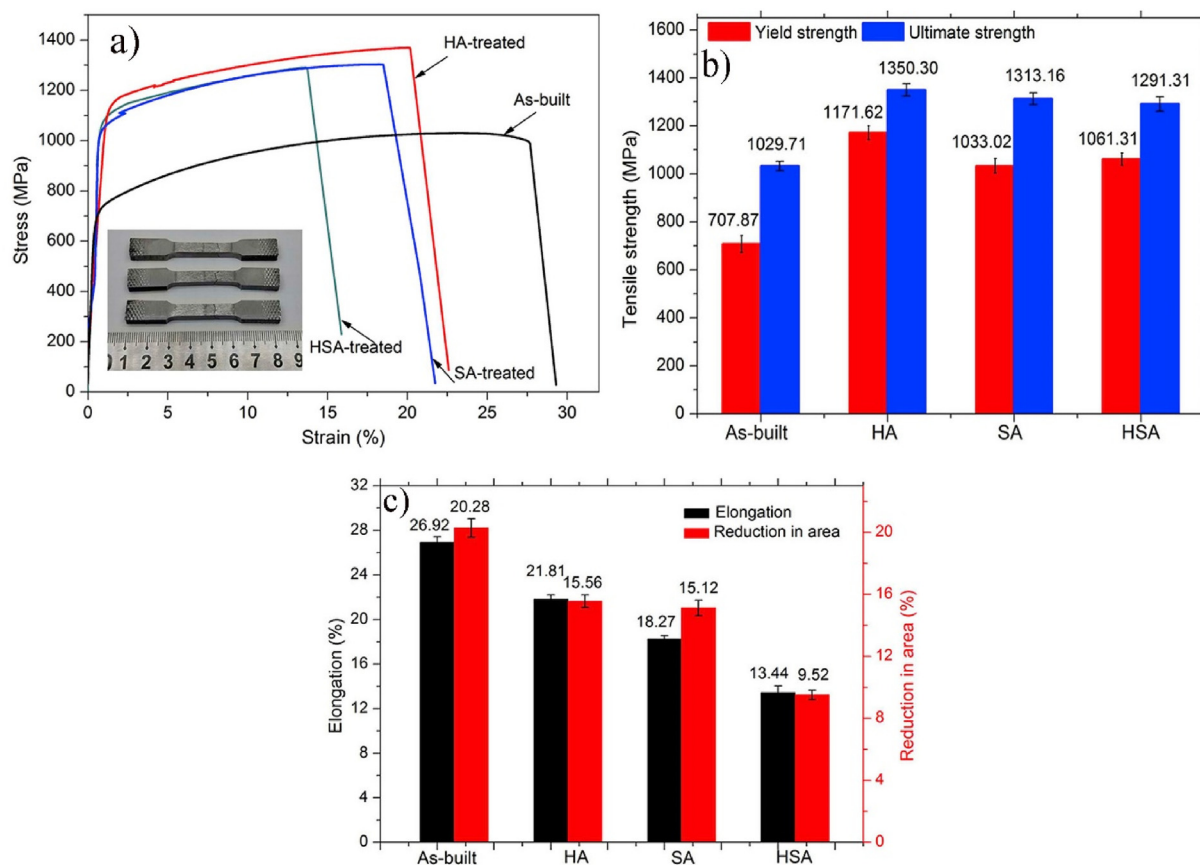


Fig. 13 – (a) Tensile stress–strain curves, (b) yield and ultimate strengths, and (c) ductility of as-built and heat-treated WC/Inconel 718 composites [218].

Currently, the available materials for additive manufacturing are rather restricted. There is an urgency in increasing the maturity of available materials to allow large scale miniature-part manufacturing. Moreover, reducing the existing gap between AM and conventional processing, such as casting and forging, is critical.

11. Microselective laser melting

Microfabrication technologies have flourished because of the ever-increasing demand for miniaturisation in numerous industries, including medicine, automotive, biotechnology, energy, communications, and optics [244]. Although metallic LPBF have already been commercialised for various industrial fields, their applicability to microscale and mesoscale fabrication has been restricted. For microscale fabrication, LPBF has only recently been employed to produce 3D microfeatures on various metal materials. The first micro-LPBF system was developed over a decade ago at Laserinstitut Mittelsachsen e.V [245]. The system uses a Q-switched neodymium-doped yttrium aluminium garnet (Nd:YAG) laser with a power range of 0.5–2 kW. The same research team has improved the technique by using two circular-sectioned rakes to spread the powder [246]. In 2013, Gieseke et al. [247] developed AISI 316 L hollow microneedles with walls as thin as 50 μm using a micro-LPBF system with a laser spot diameter of 19.4 μm . Fischer et al. [248] have described the scale features of micro-LPBFs as follows: laser beam diameter <40 μm , layer thickness <10 μm , and powder particle size <10 μm . Despite the small spot size and fine powder, the fabricated parts exhibit poor surface roughness (R_a 8 μm). The poor finish might be due to fine powder agglomeration, causing uneven powder distribution.

Christian et al. [247] have fabricated complicated helical shapes with a minimum strut diameter of 60 μm , as shown in Fig. 14(a); however, the study has reported partial strut failure. Yadroitsev and Bertrand [249] have processed stainless steel 904 L microfluidic systems using a commercial system, PM 100,

as shown in Fig. 14(b). In 2014, Fischer et al. [248] revealed the process parameters of micro-LPBF that allow 316 L steel powder layer thickness of 5–7 μm with minimum surface roughness, R_a , of 7.29 μm . Abele and Kniepkamp [250] have also investigated the effect of micro-LPBF process parameters and contour scanning on the surface quality of fabricated parts.

Roberts and Tien [251] recently (2017) designed and fabricated high-aspect-ratio, high-density stainless steel micro-electrode arrays (MEAs) via microlaser sintering from stainless steel powder with 5 and 30 μm vertical and lateral resolutions, respectively. In 2019, Roy et al. [252] developed microscale selective laser sintering ($\mu\text{-SLS}$) that can produce 3D metal parts with feature resolutions better than 5 μm and a high throughput that can be applied in a production environment. However, the undesirable heat dissipation inside the powder bed during laser sintering is the main source of error, restricting the minimum feature size of the processed components. In 2021, Grose et al. [253] introduced a method for determining the effective thermal conductivity of metal nanoparticle beds using a $\mu\text{-SLS}$ process. The group is able to produce single-micron resolution parts via microscale selective laser sintering. Nagarajan et al. [254] have comprehensively reviewed micro-LPBF and SLS for producing microscale features. From the literature mentioned above, it is clear that micro-LPBF has received much less attention than traditional macroscale LPBF.

12. Future research trends

LPBF is gaining considerable attention in the scientific community and is becoming a hotspot in the field of additive manufacturing. Identifying the challenges from previous works will assist researchers in solving the critical issues associated with LPBF processing. Thus, the tasks for future work are as follows:

1. The main challenge in LPBF is defect formation. Defects can significantly influence the part quality, hence, affecting the real-world application of LPBF. Defects can

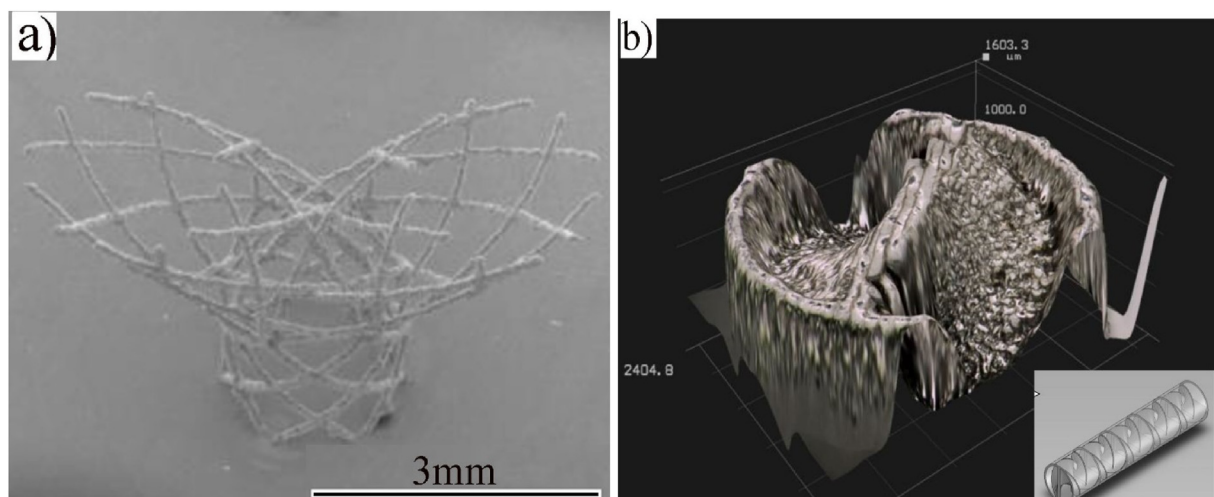


Fig. 14 – Microparts fabricated by micro-LPBF technique. (a) Y-shaped helix structures with strut diameter of 60 μm [247]. (b) Top view of microfluidic systems fabricated from stainless steel 904L [249].

easily form during LPBF because of inappropriate selection of process parameters and the complexity of defect formation. Thus, for high-quality LPBF-built parts, defect monitoring, simulation, and modelling to track the processing steps regarding stress/deformation evolution and defect initiation and control are necessary. Achieving LPBF-fabricated parts with no defects is possible.

2. Because it is difficult to directly detect complex microstructure evolution, which includes grain growth, phase transformation, texture, formation of precipitations, and dislocations, during the SLM part fabrication, computational models are effective tools for predicting the changes that occur in the developed microstructure during LPBF. The most widely utilised models in LPBF, CA, PF, and MC, have concentrated on the evolution of the microstructure during a single scan. Because the LPBF process involves remelting the previously solidified layer, continuing to develop models for multiscan/multilayer to simulate the microstructure of the LPBF-built metal parts is necessary.
3. An innovative approach in additive manufacturing that is in development is 4D printing. With time as the fourth dimension, the technique is evolving and can be applied in medicine, defence, and the aerospace sector, especially in spaces with deployable solar panels, composite hinges, and antennas. The process of 4D printing is based on 3D printing. The stimulus (water, heat, moisture, light, or a combination of both), smart or stimulus-responsive materials, interaction mechanisms, and mathematical modelling are similar. Four-dimensional printing is different from 3D printing in that it is time-dependent, printer-independent, predictable, and targets shape/property/functionality evolution [255]. To date, 4D printing has been limited to polymers and TiNi shape memory alloys [83,255], which means that smart materials still need to be developed to expand the applicability of the 4D printing technology. Thus, future interdisciplinary research studies need to be developed, optimised, and maximised for the potential use of 4D printing in engineering applications.

13. Conclusions

This paper reviews the LPBF process parameters and their effects on defect formation mechanisms and microstructure evolution in LPBF processes. In addition, newly developed micro-LPBF for 3D microfeature fabrication on metals is briefly reviewed. The following summary is drawn from the above analysis:

1. The densification and quality of LPBF-metallic parts can be enhanced by adjusting the process parameters. Nevertheless, the optimal LPBF process parameters vary according to the optimal forming areas of different equipment.
2. The most prevalent metallurgical defects in LPBF-processed metallic parts include porosity, balling, cracking, inclusions of oxides, and irregular incomplete fusion holes. These defects can adversely affect the mechanical properties of the part, limiting the application of the LPBF technique. The

process parameters in LPBF, including laser power, hatch spacing, scan speed, layer thickness, and scan strategy, significantly affect defect formation.

3. Metal additive manufacturing processes can generally produce certain characteristics in the grain structure, including both columnar and equiaxed grains, which can coexist with columnar grains or between every two layers of columnar grains. Nucleation preferentially occurs at the fusion line during solidification because of low activation energy. Epitaxial nucleation is the nucleation mechanism in LPBF.
4. Precipitation and phase transformation during LPBF is also discussed. Similar to titanium alloys, LPBF precipitation can occur with phase transformation simultaneously. The laser processing parameters and/or the chemical composition of the raw powder greatly affect the type and characteristics of the precipitates. The precipitates significantly influence the mechanical properties of as-built LPBF parts. Additionally, heat treatment, which is considered an effective technique to minimise and homogenise the generated residual stress, is reviewed.
5. Finally, this paper presents a brief overview of the newly developed micro-LPBF for fabricating 3D microfeatures on various metals. Although LPBF has already been commercialised for various industrial fields, its application has been restricted to macroscale and mesoscale fabrication. Thus, micro-LPBF deserves further attention as the field moves forward.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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